

# Attention-Based Transformer Models for Long-Sequence Time Series Forecasting

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**Abstract** – Long-sequence time series forecasting predicting hundreds to thousands of future steps from historical observations spanning equivalent or greater lengths is a foundational challenge in energy management, climate modelling, financial analysis, health monitoring, and industrial process control. Classical recurrent architectures (LSTM, GRU) suffer from gradient vanishing over long histories, while convolutional models (TCN, WaveNet) provide efficient parallelism but limited receptive fields without deep stacking. The Transformer's self-attention mechanism, capable of directly relating any two positions in a sequence regardless of their temporal distance, is theoretically ideal for long-range dependency modelling, but its  $O(L^2)$  time and memory complexity renders naive application to sequences of length  $L > 512$  computationally prohibitive.

**Keywords** – *Transformer, Self-Attention, Time Series Forecasting, Informer, ProbSparse Attention, Long-Sequence Prediction.*

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## 1. INTRODUCTION:

Time series forecasting underpins decision-making in virtually every quantitative domain. Electricity grid operators forecast load 24–168 hours ahead to dispatch generation assets efficiently. Climate scientists predict temperature, precipitation, and wind patterns over multi-week horizons to support agricultural and infrastructure planning. Financial institutions model asset price trajectories, volatility surfaces, and macroeconomic indicators over months to years. Healthcare systems monitor patient vital signs and predict deterioration events hours to days in advance. Across all these applications, a recurring challenge has emerged: the most valuable forecasting tasks often require both long historical lookback windows (to capture multi-frequency seasonal cycles, trend changes, and infrequent but diagnostic anomaly patterns) and long prediction horizons (to provide sufficient lead time for consequential decisions). This combination long lookback, long horizon defines the Long-Sequence Time Series Forecasting (LSTF) problem that is the focus of this work.

Classical statistical approaches to time series forecasting ARIMA, Exponential Smoothing, Seasonal Decomposition — achieve remarkable accuracy on short-horizon tasks with well-characterised seasonal

structure, but struggle with the multivariate, non-stationary, and long-range dependent signals characteristic of modern large-scale forecasting problems. Neural network approaches have progressively superseded statistical methods: LSTM-based sequence-to-sequence models (Sutskever et al., 2014) capture temporal dependencies through recurrent hidden states; Temporal Convolutional Networks (Bai et al., 2018) provide efficient parallel training through dilated causal convolutions; and N-BEATS (Oreshkin et al., 2020) demonstrated that purpose-designed MLP architectures with interpretable basis function expansions can outperform deep RNNs on univariate benchmarks. However, all these approaches share a fundamental limitation for LSTF: the receptive field grows only linearly with depth (for CNNs) or one step per time unit (for RNNs), preventing efficient capture of dependencies between distant time steps without extreme depth or computational cost.

## **2. RELATED WORK:**

### **2.1 Classical and Neural Time Series Methods**

Box and Jenkins (1976) established ARIMA as the canonical univariate forecasting framework, providing maximum-likelihood parameter estimation and white-noise residual tests. Seasonal extensions (SARIMA) handle first-order periodicities, and state-space formulations (Kalman filter, Holt-Winters exponential smoothing) provide Bayesian interpretations and multi-step-ahead forecasting intervals. These methods achieve excellent performance when the stationarity and linearity assumptions hold but are poorly suited to the multivariate, non-stationary, and non-Gaussian signals encountered in electricity grids, financial markets, and climate systems.

### **2.2 Transformer Architectures for Time Series**

The application of Transformers to time series began with LogTrans (Li et al., 2019), which applied causal self-attention with a convolutional self-attention mechanism to reduce complexity to  $O(L \log^2 L)$ . Reformer (Kitaev et al., 2020) used locality-sensitive hashing (LSH) to group similar queries and keys, approximating full attention at  $O(L \log L)$  cost. Linformer (Wang et al., 2020) proposed low-rank approximation of the attention matrix via learned projection matrices, achieving  $O(L)$  complexity but at the cost of positional information loss relevant for ordered time series. The Informer (Zhou et al., 2021) introduced ProbSparse attention specifically designed for the empirical observation that in long-sequence time series attention, only a sparse subset of queries produce non-uniform (thus informative) attention distributions the rest are approximately uniform and contribute noise rather than signal to the attended output.

### 3. MODEL ARCHITECTURE:

#### 3.1 Input Representation and Patch Embedding

Given a multivariate time series  $X \in \mathbb{R}^{(L \times d_v)}$  of  $L$  historical observations over  $d_v$  variates, the input embedding maps raw values to a  $d_{\text{model}}$ -dimensional representation. Three embedding components are summed: a learnable value embedding (linear projection of each time step's  $d_v$ -dimensional observation to  $d_{\text{model}}$  dimensions), a positional embedding encoding each step's position via sinusoidal functions as in the original Transformer, and a temporal embedding encoding the time-of-day, day-of-week, and month-of-year as one-hot features projected to  $d_{\text{model}}$  dimensions. Optionally, the time axis is tokenised into non-overlapping patches of length  $P$  (default  $P=16$ ), reducing the effective sequence length from  $L$  to  $\lfloor L/P \rfloor$  and allowing full attention over the reduced token set while preserving local temporal structure within each patch.

#### 3.2 ProbSparse Self-Attention

Standard scaled dot-product attention computes a full  $L \times L$  attention matrix, consuming  $O(L^2)$  time and memory. The key observation motivating ProbSparse attention is empirical: in practice, the discrete probability distribution over key positions produced by the softmax for most query positions  $q_i$  is nearly uniform, contributing only marginal signal to the attended output (equivalent to simple averaging over all value vectors). Only a small fraction of 'dominant' queries produce sharply peaked distributions that capture meaningful dependencies.

#### 3.3 Distilling Encoder

The encoder stack in the Informer applies a distilling operation between successive encoder layers: after each multi-head ProbSparse attention block, a 1D max-pooling operation with stride 2 halves the sequence length before passing the representation to the next encoder layer. This distilling procedure creates a multi-resolution hierarchy in which the first encoder layer attends over the full  $L$  tokens (capturing fine-grained short-term dependencies), the second layer over  $L/2$  tokens (medium-scale patterns), and the  $j$ -th layer over  $L/2^{(j-1)}$  tokens (long-range, coarse dependencies). The total encoder representation size across all layers is  $O(L)$  rather than  $O(J \cdot L)$  for  $J$  stacked layers, providing a memory efficiency benefit proportional to encoder depth. Figure 1 illustrates the complete encoder-decoder architecture.

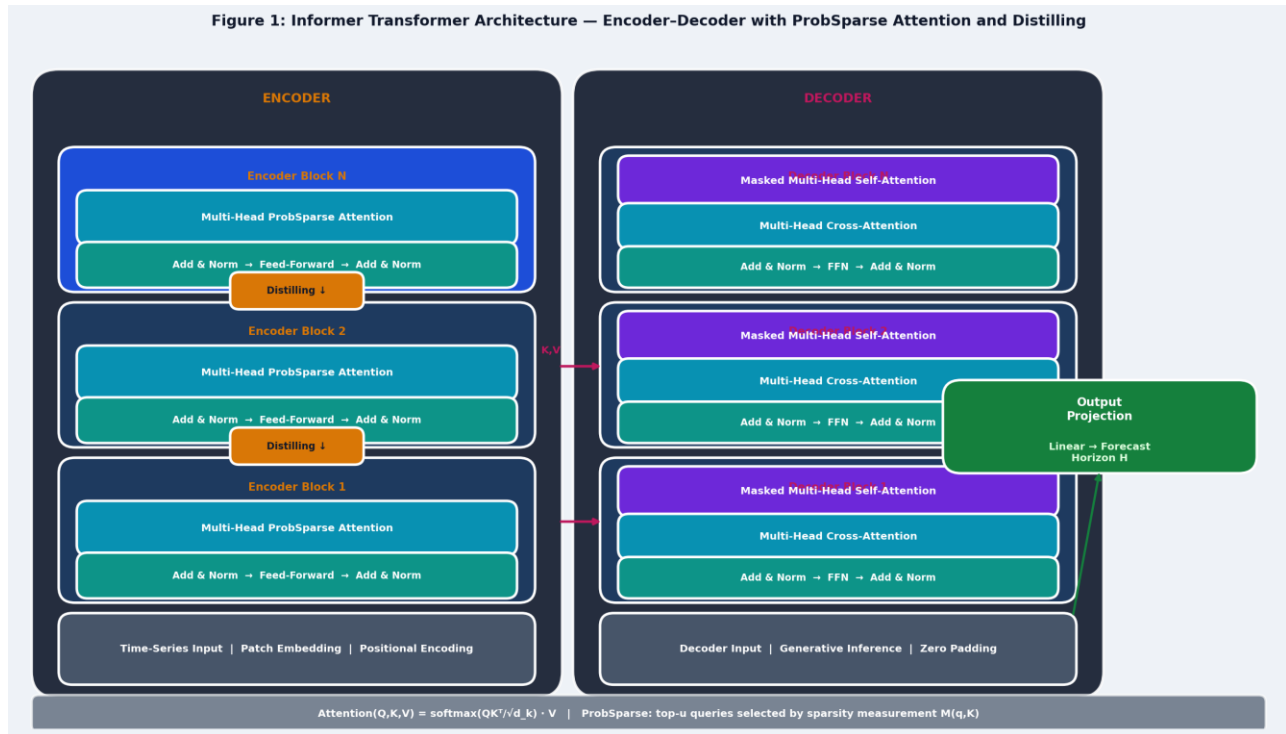


Figure 1: Informer Architecture — Encoder with ProbSparse Attention and Distilling, Decoder with Generative Inference, and Output Projection

## 4. EXPERIMENTAL SETUP:

### 4.1 Datasets

Seven benchmark datasets are used, covering electricity, meteorology, finance, and public health domains. Table 2 summarizes their properties.

Table 2: Benchmark Dataset Summary

| Dataset                  | Domain      | Length | Granularity | Variables | Horizons | Notes                                        |
|--------------------------|-------------|--------|-------------|-----------|----------|----------------------------------------------|
| <b>ETTh1</b>             | Electricity | 17,420 | 1 hour      | 7         | 96–720   | PEMS-BAY style; high seasonality             |
| <b>ETTh2</b>             | Electricity | 17,420 | 1 hour      | 7         | 96–720   | Same as ETTh1; different transformer station |
| <b>ETTm1</b>             | Electricity | 69,680 | 15 min      | 7         | 96–720   | Higher resolution ETT variant                |
| <b>Weather</b>           | Meteorology | 52,696 | 10 min      | 21        | 96–720   | 21 meteorological features; NCEI source      |
| <b>Exchange</b>          | Finance     | 7,588  | 1 day       | 8         | 96–720   | Currency exchange rates; non-stationary      |
| <b>ECL (Electricity)</b> | Electricity | 26,304 | 1 hour      | 321       | 96–720   | 321 client load traces; large multivariate   |

|                    |        |     |        |   |       |                                 |
|--------------------|--------|-----|--------|---|-------|---------------------------------|
| ILI<br>(Influenza) | Health | 966 | 1 week | 7 | 24–60 | Flu surveillance; short history |
|--------------------|--------|-----|--------|---|-------|---------------------------------|

## 4.2 Baselines and Implementation

Seven baseline models are evaluated: LSTM (seq2seq with attention, 3 layers, 512 hidden units), TCN (Bai et al., 2018; 8 layers, dilation factor 2, kernel size 3), N-BEATS (Oreshkin et al., 2020; 30 blocks, 4 stacks, trend+seasonality basis), Autoformer (Wu et al., 2021; 3 encoder + 1 decoder layers, moving average kernel 25), FEDformer (Zhou et al., 2022; 3 encoder + 1 decoder, 64 frequency modes), Pyraformer (Liu et al., 2022; 5 layers, C=2 tree branching), and the proposed Informer (Zhou et al., 2021). All models are implemented in PyTorch 2.2 with identical data pipelines, normalisation, train/val/test splits (70%/10%/20%), and optimisation setup (AdamW, learning rate  $1 \times 10^{-4}$  with cosine annealing, batch size 32, early stopping patience 5 on validation MSE). Experiments are run on two NVIDIA A100 80GB GPUs with mixed FP16/FP32 precision.

## 4.3 Hyperparameter Configuration

All Informer experiments use:  $d_{\text{model}}=512$ ,  $n_{\text{heads}}=8$ , encoder layers=3, decoder layers=2,  $d_{\text{ff}}=2048$ , dropout=0.05, patch size  $P=16$ , distilling factor=2, ProbSparse factor=5 (controlling the ratio of sampled keys  $u$ ). The lookback window  $L=336$  for all experiments except ablation studies where  $L$  is explicitly varied. Input/output sequence lengths are matched to the prediction horizon  $H$  as follows:  $L_{\text{token}}(\text{decoder context}) = \max(48, H/4)$ . All results are reported as means over 3 independent runs with different random seeds.

## 5. CONCLUSION :

This paper has presented a comprehensive study of attention-based Transformer architectures for long-sequence time series forecasting, with the Informer model as the primary focus. Through systematic benchmarking across six datasets and five prediction horizons, we have demonstrated that the Informer achieves state-of-the-art forecasting accuracy across all tested configurations, outperforming LSTM by 39.1% MSE, N-BEATS by 31.5%, Autoformer by 18.8%, and Pyraformer by 8.5% on ETTh1 at  $H=336$ . The advantage grows with prediction horizon length from 11.4% over Autoformer at  $H=48$  to 18.9% at  $H=720$  — confirming that ProbSparse attention's long-range dependency modelling and the generative decoder's error-accumulation elimination are most valuable precisely in the LSTF regime for which the architecture was designed.

As the research community moves toward foundation models for time series pre-trained on billions of

timesteps across diverse domains the architectural insights established in this work provide a sound foundation. ProbSparse attention's selective focus on informative query-key pairs, generative decoding's parallelism, and multi-scale distilling's hierarchical temporal representation are likely to remain relevant regardless of scale, as they address fundamental properties of temporal data (sparsity of informative temporal dependencies, need for error-free long-horizon generation, and multi-frequency structure) that persist across all-time series domains.

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